

ECE 330 Exam 1: Spring 2023
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 90 Minutes

NAME _____

NetID (please print) _____

USEFUL INFORMATION

$$\sin(x) = \cos(x - 90^\circ)$$

$$0 < \theta < 180^\circ \text{ (lag)}$$

$$-180^\circ < \theta < 0 \text{ (lead)}$$

$$\bar{V} = \bar{Z}\bar{I} \quad \bar{S} = \bar{V}\bar{I}^* = P + jQ$$

$$I_L = \sqrt{3}I_\phi \text{ (delta)}$$

$$V_L = \sqrt{3}V_\phi \text{ (wye)}$$

$$\bar{S}_{3\phi} = \sqrt{3}V_L I_L \angle \theta$$

$$\bar{Z}_Y = \bar{Z}_\Delta / 3$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

ABC phase sequence has A at 0, B at -120°, and C at +120°

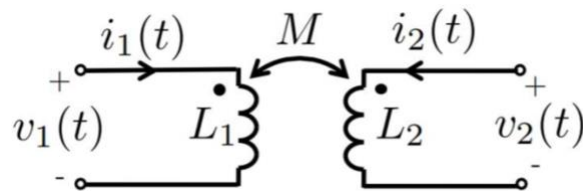
$$\int \underline{H} \cdot \underline{dl} = \int \underline{J_f} \cdot \hat{n} dA \quad \int \underline{E} \cdot \underline{dl} = -\frac{d}{dt} \left(\int \underline{B} \cdot \hat{n} dA \right) \quad \mathcal{R} = \frac{l}{\mu A} \quad Ni = \mathcal{R}\phi$$

$$\phi = BA$$

$$\lambda = N\phi = Li \text{ (if linear)}$$

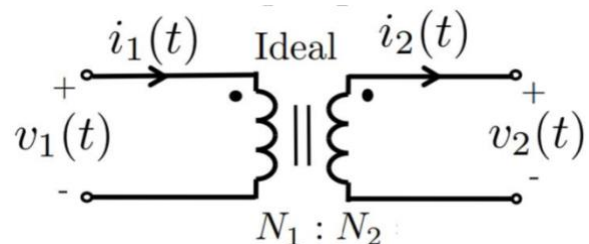
$$v = \frac{d\lambda}{dt} \quad k = \frac{M}{\sqrt{L_1 L_2}}$$

$$1 \text{ hp} = 746 \text{ W}$$



$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

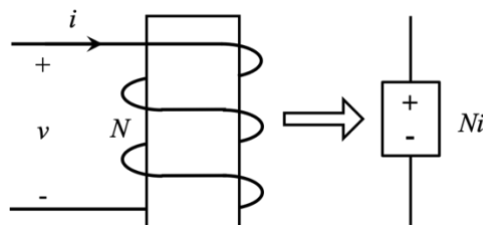
$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$



$$a = \frac{N_1}{N_2}$$

$$N_1 i_1 = N_2 i_2$$

$$\frac{v_1}{v_2} = \frac{N_1}{N_2}$$



Problem 1 (25 Points)

A $2\angle 0^\circ$ kV, 60 Hz supply powers 2 single phase loads connected in parallel. The total current supplied to the 2 loads is 100 A at a lagging power factor. Both loads consume a total power of 150 kW, and load 1 consumes 80 kVA at a PF=0.95 lagging.

- a) What is the angle of the current supplied to the 2 loads?

$$P = S(PF)$$

$$S = VI$$

$$P = VI(PF)$$

$$PF = \frac{P}{VI}$$

$$PF = \frac{150 \text{ kW}}{2(100) \text{ kVA}}$$

$$PF = \frac{150}{200} \Rightarrow \underline{\underline{PF = 0.75 \text{ lag}}}$$

$$\theta = \cos^{-1}(PF) \text{ [deg]}$$

$$\theta = \theta_v - \theta_i$$

$$\theta_v = 0$$

$$-\theta_i = \cos^{-1}(PF)$$

$$\theta_i = -\cos^{-1}(PF)$$

$$\theta_i = -\cos^{-1}(0.75)$$

$$\boxed{\theta_i = -41.41^\circ = -0.7227 \text{ rad.}}$$

$\theta_i =$ _____

- b) What is the complex power consumed by load 2?

$$\bar{S}_{tot} = \bar{V} \bar{I}^*$$

$$= VI \angle \theta$$

$$= 2(100) \angle 41.41^\circ$$

$$= 200 \angle 41.41^\circ$$

$$= 150 + j132.29 \text{ kVA}$$

$$\bar{S}_1 = 80 \angle \cos^{-1}(0.95)$$

$$\bar{S}_1 = 80 \angle 18.19^\circ \text{ kVA}$$

$$= 76 + j24.97 \text{ kVA}$$

$$\bar{S}_{tot} = \bar{S}_1 + \bar{S}_2$$

$$\bar{S}_2 = \bar{S}_{tot} - \bar{S}_1$$

$$\bar{S}_2 = (150 + j132.29) - (76 + j24.97)$$

$$\bar{S}_2 = 74 + j107.32 \text{ kVA}$$

$$= 130.36 \angle 55.41^\circ \text{ kVA}$$

$\bar{S}_2 =$ _____

- c) What is the impedance of a capacitive load that needs to be added in parallel to the 2 loads to achieve an overall power factor of $PF=0.975$ lagging? (You may state your answer either in rectangular or polar form)

$$PF_n = 0.975$$

$$\theta_n = 12.84^\circ$$

$$Q_n = P_o \tan(\theta_n)$$

$$Q_n = 150 \tan(12.84^\circ)$$

$$Q_n = 34.19 \text{ kVAR}$$

$$Q_c = Q_n - Q_o$$

$$= 34.19 - 132.29$$

$$Q_c = -98.1 \text{ kVAR}$$

$$\bar{S}_c = 0 - j98.1 \text{ kVAR}$$

$$= 98.1 \angle -90^\circ \text{ kVAR}$$

$$\bar{S}_c = \frac{V^2}{\bar{Z}_c^*}$$

$$\bar{Z}_c = \left(\frac{V^2}{\bar{S}_c} \right)^*$$

$$= \left(\frac{2000^2}{98.1 \times 10^3 \angle -90^\circ} \right)^*$$

$$\bar{Z}_c = 40.77 \angle -90^\circ \Omega$$

$$= 0 - j40.77 \Omega$$

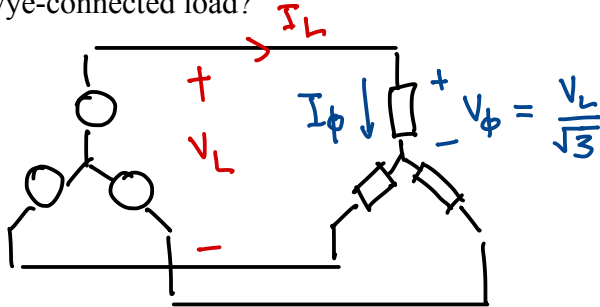
$$\bar{Z}_{cap} = \underline{\hspace{10cm}}$$

Problem 2 (25 Points)

A balanced 3-phase, 3 wire, 60 Hz, wye-connected source is serving a wye-connected load of unknown total power.

The three-phase line voltage and line current measured at the source are 578 V and 30 A respectively. You may ignore the impedance of the wires connecting the source to the load.

- a) What is the per-phase voltage and per-phase current magnitudes measured at the unknown wye-connected load?

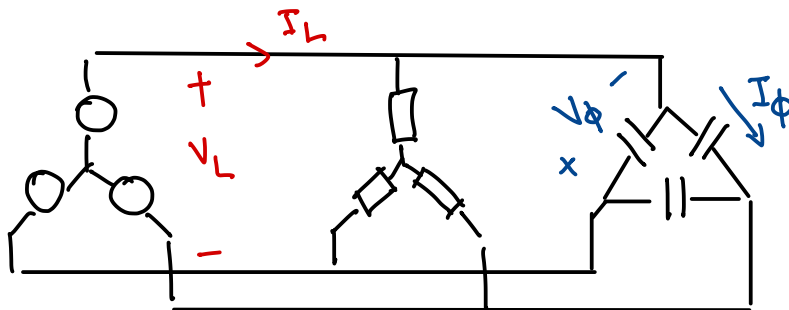


$$V_{\phi} = \frac{578}{\sqrt{3}} \text{ V}$$

$$I_{\phi} = 30 \text{ A}$$

In attempt to reduce the line current at the source, 30 kVAR of 3-phase delta-connected capacitance is added in parallel to the loads

- b) What is the per-phase voltage and per-phase current magnitudes measured at the capacitor?



$$\text{Each phase : } 10 \text{ kVAR, } 578 \text{ V}$$

$$\text{Current : } 10 \text{ k} / 578 = 17.301 \text{ A}$$

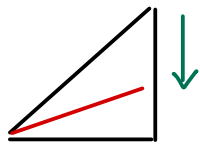
$$V_{\phi} = 578 \text{ V}$$

$$I_{\phi} = 17.301 \text{ A}$$

With the capacitance added, the line current at the source *increases* to 45 A, instead of decreasing as we had intended.

- c) What is the total three-phase complex power of the unknown wye-connected load? Please take care to indicate whether the load is capacitive or inductive, and state your units. (You may state your answer either in rectangular or polar form)

Capacitive load. ✓ current does increase.



Inductive
load

X
Current
should
decrease

$$|\bar{S}_{old}| = \sqrt{3}(578)(30) = 30.034 \text{ kVA} \approx 30 \text{ kVA}$$

$$Q_{cap} = 30 \text{ kVAR}$$

$$|\bar{S}_{new}| = \sqrt{3}(578)(45) = 45.050 \text{ kVA} \approx 45 \text{ kVA}$$

$$S_{old}^2 = P^2 + Q^2$$

$$S_{new}^2 = P^2 + (Q + Q_{cap})^2$$

$$S_{new}^2 - S_{old}^2 = (Q + Q_{cap})^2 - Q^2$$

$$= \cancel{Q^2} + 2Q_{cap}Q + Q_{cap}^2 - \cancel{Q^2}$$

$$Q = \frac{S_{new}^2 - S_{old}^2 - Q_{cap}^2}{2Q_{cap}},$$

$$= (45^2 - 30^2 - 30^2) / (-2 \times 30)$$

$$= -3.75 \text{ VAR}$$

$$P = \sqrt{S_{old}^2 - Q^2}$$

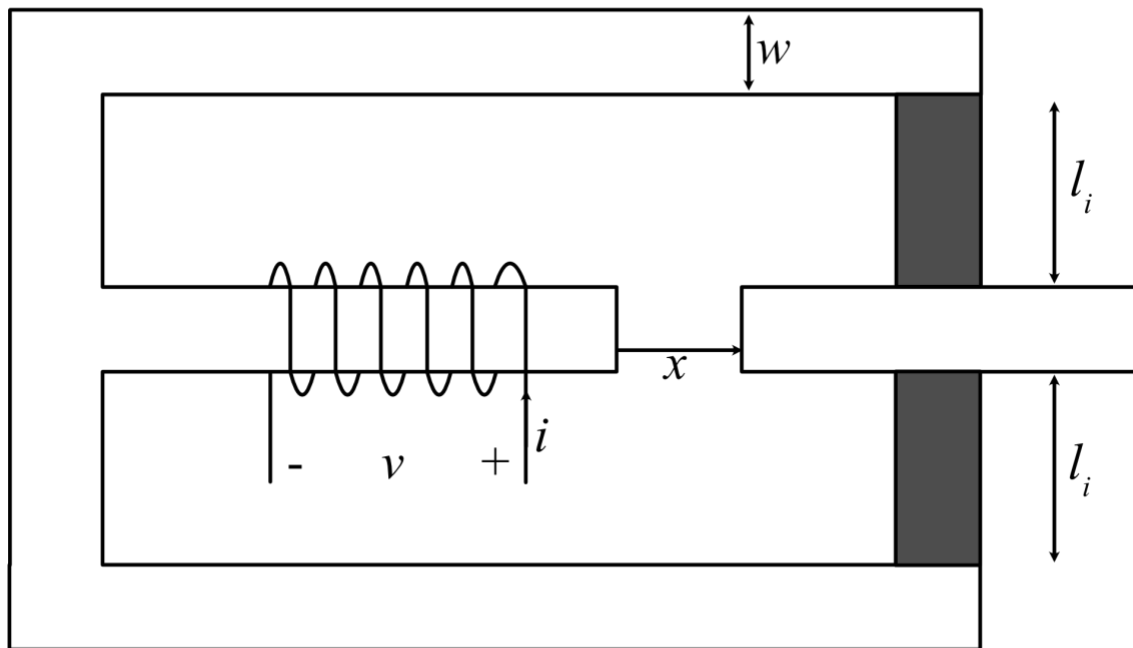
$$= 29.7647$$

CAPACITIVE

INDUCTIVE

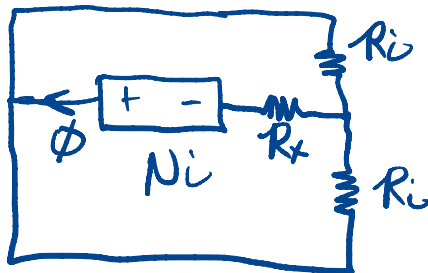
$$\bar{S}_{Load} = 29.765 - j3.75 \text{ kVA}$$

Problem 3 (25 Points)



An infinitely permeable iron core given above with width $w=2$ cm and depth into the page $d=3$ cm has a coil with 450 turns wrapped around it as shown. The core contains two inserts with a permeability of $\mu_i=2450\mu_0$ with a length $l_i=10$ cm. The width of the inserts and the depth into the page of the inserts are also $w=2$ cm and $d=3$ cm. The core also contains an air gap with $x=20$ mm. Fringing effects can be neglected.

- a) Draw the magnetic equivalent circuit for this system. Include flux direction, MMF polarity, and numerical values for reluctances.



$$R_i = \frac{l_i}{\mu_i A_i} = \frac{10 \text{ cm}}{(2450\mu_0)(6 \text{ cm}^2)}$$

$$R_i = 5.413 \times 10^4 \text{ At/Wb}$$

$$R_x = \frac{x}{\mu_0 A} = \frac{20 \text{ mm}}{\mu_0 (6 \text{ cm}^2)}$$

$$R_x = 2.653 \times 10^7 \text{ At/Wb}$$

- b) What is the numerical value for the total flux linkage of the coil written in terms of the unknown current $i(t)$?

$$Ni = R_{eq} \phi$$

$$\phi = \frac{Ni}{R_{eq}}$$

$$R_{eq} = R_x + \left(\frac{1}{R_L} + \frac{1}{R_L} \right)^{-1}$$

$$R_{eq} = R_x + \frac{R_L}{2}$$

$$R_{eq} = 2.656 \times 10^{-7} \text{ At/Wb}$$

$$\lambda = Ni$$

$$\lambda = \frac{N^2}{R_{eq}} i(t)$$

$$\lambda = \frac{450^2}{2.656 \times 10^{-7}} i(t)$$

$$\boxed{\lambda = 0.007624 i(t)}$$

$\lambda =$ _____

- c) If the current into the coil is given as a function of time as $i(t) = 5(1 - e^{-4t})$, what is the numerical value for the voltage across the coil $v(t)$ as a function of time t ?

$$v(t) = \frac{d\lambda}{dt}$$

$$v(t) = \frac{d}{dt} (0.007624 i(t))$$

$$v(t) = 0.007624 \frac{di}{dt}$$

$$i(t) = 5(1 - e^{-4t})$$

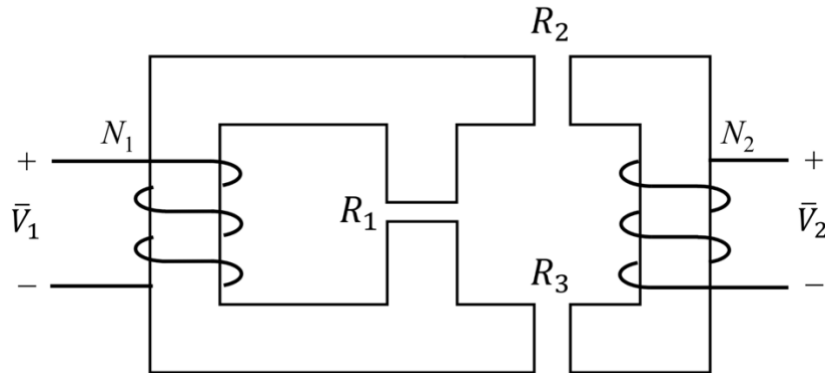
$$\frac{di}{dt} = 20e^{-4t}$$

$$\Rightarrow \boxed{v(t) = 0.1525 e^{-4t}}$$

$v(t) =$ _____

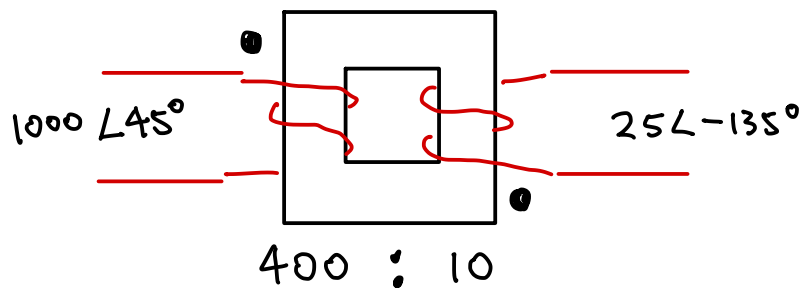
Problem 4 (25 Points)

Your firm is building a transformer, comprising two coils that are wound to have $N_1 = 400$ turns and $N_2 = 10$ turns respectively. The three air gaps are sized to have reluctances of $R_1 = 10 \text{ At/Wb}$, $R_2 = 500 \text{ At/Wb}$, and $R_3 = 500 \text{ At/Wb}$ as labeled. You may treat the permeability of the magnetic core as infinite.



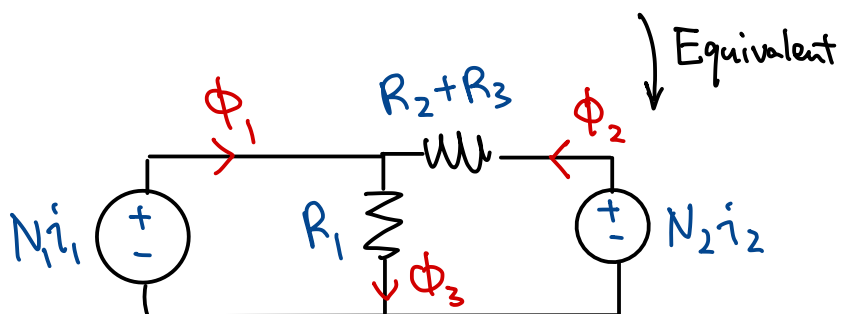
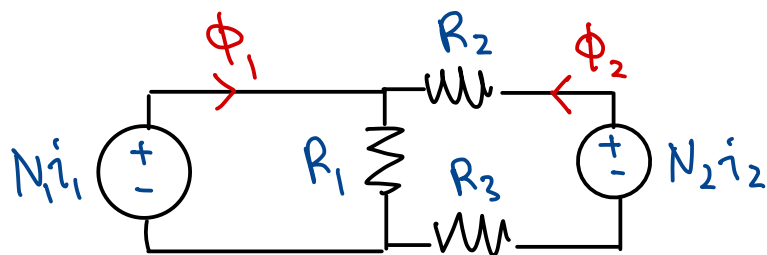
- a) Assuming for now that the transformer is ideal, what would the output voltage $\bar{V}_2$ be at open circuit with an input voltage $\bar{V}_1 = 1 \angle 45^\circ \text{ kV (rms)}$ at 60 Hz? (Please specify both the phase and the magnitude of the output voltage $\bar{V}_2$, noting that the magnitude must be a nonnegative number, and briefly justify your answer.)

Ideally:



$$\bar{V}_2 = \underline{25 \angle -135^\circ \text{ V}}$$

- b) Compute the self inductances L_1 , L_2 , and the mutual inductance M of the two coils.



$$\phi_2 = \frac{N_2 i_2 - N_1 i_1}{R_2 + R_3}, \quad \phi_3 = \frac{N_1 i_1}{R_1}$$

$$\phi_1 + \phi_2 = \phi_3 \Rightarrow \phi_1 = \phi_3 - \phi_2$$

$$= \frac{N_1 i_1}{R_1} - \left(\frac{N_2 i_2 - N_1 i_1}{R_2 + R_3} \right)$$

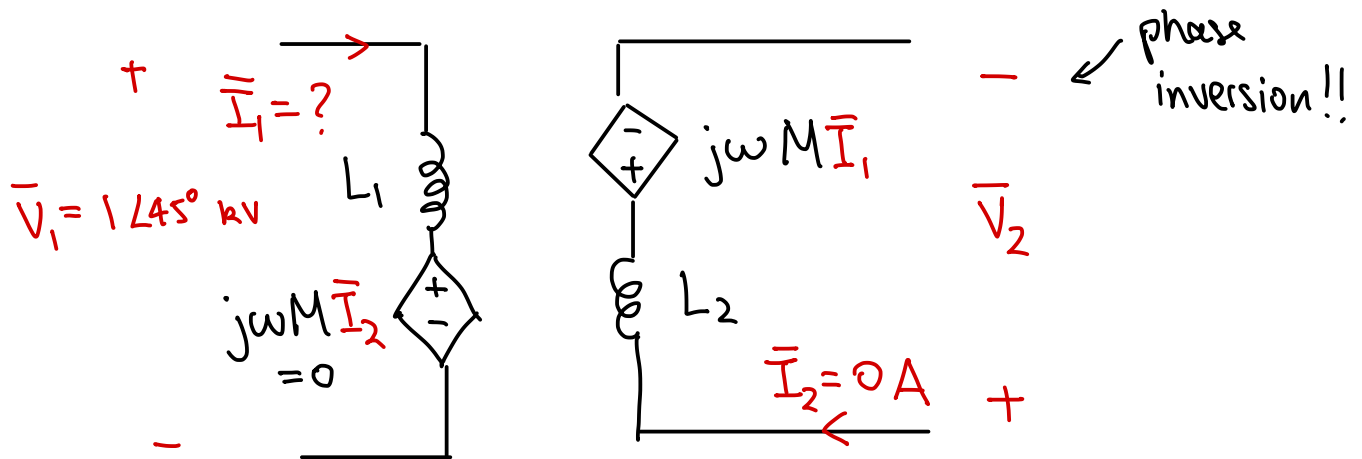
$$= \left(\frac{1}{R_1} + \frac{1}{R_2 + R_3} \right) N_1 i_1 - \frac{1}{R_2 + R_3} N_2 i_2$$

$$L_1 = \frac{N_1^2 [R_1^{-1} + (R_2 + R_3)^{-1}]}{1} = 16160 \text{ H}$$

$$L_2 = \frac{N_2^2 (R_2 + R_3)^{-1}}{1} = 0.1 \text{ H}$$

$$M = \frac{N_1 N_2 (R_2 + R_3)^{-1}}{1} = 4 \text{ H}$$

- c) Using the self and mutual inductance values you have just computed, what is the actual output voltage $\bar{V}_2$ at open circuit with an input voltage $\bar{V}_1 = 1\angle 45^\circ$ kV (rms) at 60 Hz? (Please specify both the phase and the magnitude of the output voltage $\bar{V}_2$, noting that the magnitude must be a nonnegative number, and briefly justify your answer.)



$$\bar{V}_1 = j\omega L_1 \bar{I}_1 + \cancel{j\omega L_2 \bar{I}_2}^0$$

$$\bar{V}_2 = j\omega M \bar{I}_1 + \cancel{j\omega L_2 \bar{I}_2}^0$$

$$\bar{V}_2 = \frac{j\omega M}{j\omega L_1} \bar{V}_1 = \frac{M}{L_1} \bar{V}_1 = \frac{4}{16160} \cdot 1000 \angle 45^\circ$$

finally, adjust for phase inversion.

$$\bar{V}_2 = \underline{0.2475 \angle -135^\circ} \quad \text{V}$$